

05/06/26 DAY 10 "Kamineni Dhana Sri Keerthi"

### Working on

01. Yesterday presentation questions.
02. Matlab code for RLRC eq's for a defect free TSV
03. Matlab code for RLRC eq's for a detected TSV
  - 3.1 RL model
  - 3.2 Leak & Leak model.

### Q1 Presentation questions

#### 1.1 why we use distributed model

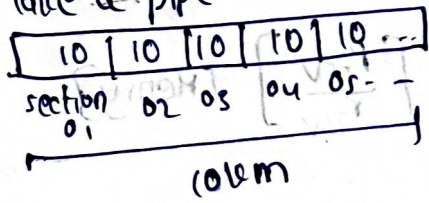
First analogous example

→ Take a water pipe

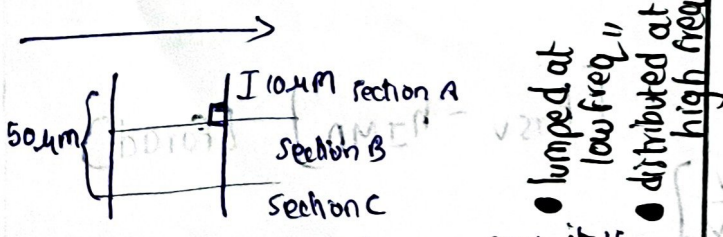
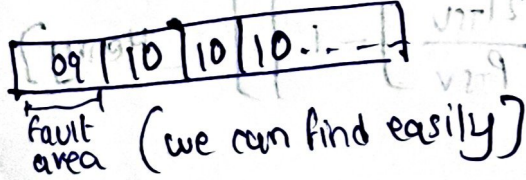


→ There's fault and it became  $99\Omega$  now we don't know where the leak is in the 10km (This is lumped) ↗

↗ Take a pipe



if a leak is there



distributed model can say it is from section A but lumped model cant say whether it is 10μm down or 25μm down.

### 1.2 What is leakage signal / current

A small unwanted current that flows through a material [insulator] which should supposed to block it.

#### 1.3 why si sub?

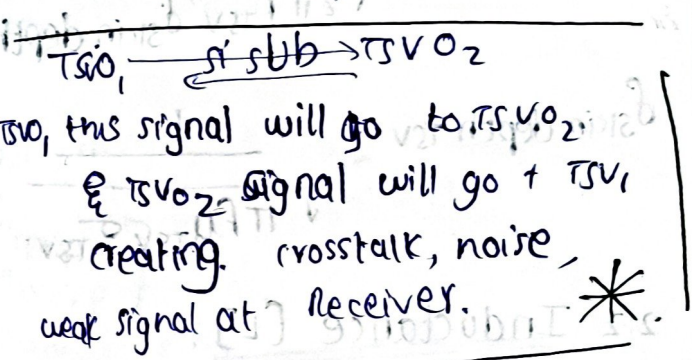
↗ it is base & its the place where we place transistors.

#### 1.4 why don't we use insulator instead of si sub?

↗ Transistors need silicon to operate.

#### 1.5 why insulating coat btw si sub & TSV

↗ TSV supposed to carry no si sub one chip to upper chip.



→ leakage current → dissipation of P → heat

#### 1.6 Are all insulators in TSV same?

↗ why we are using diff. types of insulators?

↗ at place where what property we need.

↗ Prevent leakage TSV to si (SiO<sub>2</sub>) because of its high resistivity & Thermal stability

↗ we need to separate two bumps (metals) we use IMD because intermetal dielectric

↗ have low capacitance. so we use this here instead of SiO<sub>2</sub>

|                                                            |                                                                       |
|------------------------------------------------------------|-----------------------------------------------------------------------|
| Prevent leakage → ↑ resistivity                            | ↓ delay - low K dielectric                                            |
| IMD can handle thermal stress while SiO <sub>2</sub> cant. | under fill can protect the device from moisture than SiO <sub>2</sub> |



## 2 Matlab code for a defect free TSV model

Formulae  $R, L, G, C$

### 2.1 Resistance:

$$R_{TSV} = \sqrt{[R_{dc,TSV}]^2 + [R_{ac,TSV}]^2} \quad (\Omega)$$

where

$$R_{dc,TSV} = \rho_{TSV} \frac{h_{TSV}}{\pi [r_{TSV}]^2} \quad \Omega$$

$$R_{ac,TSV} = k_p \left( \frac{\rho_{TSV} h_{TSV}}{2\pi r_{TSV} \delta_{skin\ depth\ TSV} - \pi \delta_{skin\ depth\ TSV}^2} \right) [\Omega]$$

$$\delta_{skin\ depth\ TSV} = \frac{1}{\sqrt{\pi f \mu_{TSV} \sigma_{TSV}}} \quad (cm)$$

### 2.2 Inductance [L]:

#### 2.2.1 Self Inductance

$$L_{TSV} = \frac{1}{2} \left[ \frac{\mu_0 \mu_{r,TSV} h_{TSV}}{2\pi} \right] \ln \left[ \frac{\rho_{TSV}}{r_{TSV}} \right] \quad [Henry]$$

#### 2.2.2 Mutual Inductance

$$L_M = \left[ \frac{\mu_0 \mu_{r,TSV}}{2\pi} \right] \left[ \ln \left[ \left[ \frac{2l_{TSV}}{\rho_{TSV}} \right] - 1 \right] \right] \quad [Henry]$$

### 2.3 Conductance [G]

$$G_{Si\ sub} = \frac{\pi \sigma_{Si}}{\cosh^{-1} \left[ \frac{\rho_{TSV}}{d_{TSV}} \right]} \times [h_{TSV} - h_{IMD}] \quad [Farad]$$



## 2.4 Capacitance.

$$01 \quad C_{\text{Insulator}} = \frac{1}{2} \left[ 2\pi \epsilon_0 \epsilon_{r, \text{Insulator}} \times \frac{h_{\text{TSV}} - h_{\text{IMD}}}{\ln \left[ \frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right]} \right] \text{ [Farad]}$$

$$02 \quad C_{\text{Bump01}} = \frac{\epsilon_0 \epsilon_{r, \text{IMD}} \pi \left[ (r_{\text{bump}})^2 - (r_{\text{TSV}} + t_{\text{ox}})^2 \right]}{h_{\text{IMD}}} \text{ [Farad]}$$

$$03 \quad C_{\text{Bump02}} = \frac{\epsilon_0 \epsilon_{r, \text{ox}} \pi \left[ (r_{\text{bump}})^2 - (r_{\text{TSV}} + t_{\text{ox, bot}})^2 \right]}{t_{\text{ox, bot}}} \text{ [F]}$$

$$04 \quad C_{\text{Underfill}} = \frac{\pi \epsilon_0 \epsilon_{r, \text{underfill}} h_{\text{Bump}}}{\cosh^{-1} \left[ \frac{r_{\text{TSV}}}{d_{\text{bump}}} \right]} \text{ [Farad]}$$

$$05 \quad C_{\text{IMD}} = \frac{\pi \epsilon_0 \epsilon_{r, \text{IMD}} h_{\text{IMD}}}{\cosh^{-1} \left[ \frac{r_{\text{TSV}}}{d_{\text{TSV}}} \right]} \text{ [F]}$$

$$06 \quad C_{\text{Bottom}} = \frac{\pi \epsilon_0 \epsilon_{r, \text{ox, bot}} t_{\text{ox, bot}}}{\cosh^{-1} \left[ \frac{r_{\text{TSV}}}{d_{\text{TSV}}} \right]} \text{ [F]}$$

$$07 \quad C_{\text{Si, sub}} = \frac{\pi \epsilon_0 \epsilon_{r, \text{Si}} [h_{\text{TSV}} - h_{\text{IMD}}]}{\cosh^{-1} \left[ \frac{r_{\text{TSV}}}{d_{\text{TSV}}} \right]} \text{ [F]}$$

| Symbol                                                        | Value                         | Symbol | Value. |
|---------------------------------------------------------------|-------------------------------|--------|--------|
| <u>01</u> d <sub>TSV</sub>                                    | 50 μm                         |        |        |
| <u>02</u> h <sub>TSV</sub>                                    | 100 μm                        |        |        |
| <u>03</u> p <sub>TSV</sub>                                    | 250 μm                        |        |        |
| <u>04</u> t <sub>ox</sub>                                     | 0.5 μm                        |        |        |
| <u>05</u> d <sub>bump</sub>                                   | 100 μm                        |        |        |
| <u>06</u> h <sub>bump</sub>                                   | 20 μm                         |        |        |
| <u>07</u> h <sub>IMD</sub>                                    | 2 μm                          |        |        |
| <u>09</u> t <sub>ox, bot</sub>                                | 0.5 μm                        |        |        |
| <u>10</u> σ <sub>Si</sub>                                     | 10 [S/m]                      |        |        |
| <u>11</u> ρ <sub>TSV</sub>                                    | 1.68 × 10 <sup>-8</sup> [Ω-m] |        |        |
| <u>12</u> ρ <sub>bump</sub>                                   | 1.68 × 10 <sup>-8</sup> [Ωm]  |        |        |
| <u>13</u> ε <sub>r, si</sub>                                  | 11.9                          |        |        |
| <u>14</u> ε <sub>r, IMD</sub> <small>(which material)</small> | 04 SiO <sub>2</sub>           |        |        |
| <u>15</u> ε <sub>r, underfill</sub>                           | 7 High dielectric composites  |        |        |
| <u>16</u> μ <sub>r, TSV</sub>                                 | 01                            |        |        |
| <u>17</u> μ <sub>r, bump</sub>                                | 01                            |        |        |
| <u>18</u> ε <sub>r, ox</sub>                                  | 04 SiO <sub>2</sub>           |        |        |

TSV 25 μm  
→ 50 μm

fine paper.

changed



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Code format: Mfile

P<sub>1</sub> clc;  
clear;  
close all;

P<sub>2</sub> %%%  
% constant  
%%%

P<sub>3</sub> %%%  
% Input parameters  
%%%

P<sub>4</sub> %%%  
% Equation [i].....  
%%%

P<sub>5</sub> %%%  
% Display Results  
%%%

code:

clc;  
clear;  
close all;

%%%

% Constants

%%%

$$\epsilon = 8.85 \times 10^{-12}$$

$$\epsilon_{ps} = 8.85 \times 10^{-12}; \quad \mu_0 = 4\pi \times 10^{-7}$$

$$\mu_0 = 4\pi \times 10^{-7};$$

TSV Geometry

$$d_{TSV} = 50 \times 10^{-6}$$

$$r_{TSV} = d_{TSV}/2;$$

$$h_{TSV} = 100 \times 10^{-6}$$

Pitch

$$p_{TSV} = 250 \times 10^{-6};$$

code is in matlab file.

OIP Defect Free Code

Modeling & simulation

- 1 Paper find 25um to 50um. ■
- 2 write down 01 to 09 values ■
- 3 write material names IMD underfill ■
- 4 fill what are constants and all up in code format. ■
- 5 Get code ■

① code →

parameters

modelling and simu.

②

paper → Sir.

Shivangi chandekar

Vijaya rao kumbhare  
Maya chandekar.

only introduction.

Defected TSV

[Main Pin hole defect]

Pin hole defect

in RL terms

$$R_{\text{sh}} = \frac{\rho_T t_{\text{ins}}}{L_s^2}$$

$$\rho_T = \rho_0 [1 + \alpha (T - T_0)]$$

$$t_{\text{ins}} = t_{\text{ox}}$$

$L_s$  = length of pin hole. [Choose  $L_s$  according to the take values of TSV]

$$L_{\text{sh}} = \mu_0 \mu_{\text{TSV}} t_{\text{ins}} \left[ \cosh^{-1} \left[ \frac{t_{\text{ins}}}{\sqrt{L_s^2 + t_{\text{ins}}^2}} \right] - \sqrt{2 t_{\text{ins}}^2 + L_s^2} + \frac{t_{\text{ins}}}{a} + \sqrt{L_s^2 + t_{\text{ins}}^2} \right]$$

In  $G_{\text{leak}}$  and  $G_{\text{leak}}$ .

$$C_{\text{leak}} = \left[ \frac{2\pi \epsilon_0 \epsilon_{\text{rox}} l_{\text{TSV}}}{\ln \left[ \frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right]} \right]$$

where

$$J = \frac{\omega h_{\text{pin}}}{2\pi r_{\text{TSV}}}$$

$h_{\text{pin}}$  (choose  $h_{\text{pin}}$  according to the considered dimensions)

$$G_{\text{leak}} = 2\pi \epsilon_0 \epsilon_{\text{ox}}$$

$$G_{\text{leak}} = 2\pi \sigma_s$$

$$\ln \left[ \frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right]$$

Flow

01 We are taking

02 Defect free

03 Pin hole de

Defect free

Defect 01

Defect 02

$$C_{\text{leak}} = \left[ \frac{2\pi \epsilon_0 \epsilon_{\text{air}} J r_{\text{TSV}}}{\ln \left[ \frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right]} \right]$$



$$G_{\text{leak}} = 2\pi \epsilon_0 \epsilon_{\text{oxe}}$$

$$G_{\text{leak}} = \frac{2\pi \sigma_{\text{SiO}_2} l_{\text{TSV}}}{\ln \left[ \frac{r_{\text{TSV}} + t_{\text{oxe}}}{r_{\text{TSV}}} \right]} - \frac{4\pi^2 j f \epsilon_0 \epsilon_{\text{rail}} r_{\text{TSV}}}{\ln \left[ \frac{r_{\text{TSV}} + t_{\text{oxe}}}{r_{\text{TSV}}} \right]}$$

Flow

- 01 We are taking 50um TSV
- 02 Defect free RLC equations for TSV
- 03 Pin hole defected TS Model
  - RL model [square ; Cu]
  - Cleatic Gileatic model [cylindrical ; Air]

Defect free

Defect 01

Defect 02

Analytical model [Matlab]

Validation

Hfss simulation.

1 Choose TSV dimensions



2 For defect 01

[cylindrical; Cu]



[ $\epsilon_r$  Cu]

defected 02

[cylindrical Air]

[ $\epsilon_r$  air]

3 Calculate

RLGC 01



[RL]

RLGC 02

[GC]

4 Compare delay, crosstalk, noise parameters.



5 Build defect 01

defected 02 - HFSS



6 Get results of delay



7 Compare it with matlab vs HFSS.



8 Presentation.